

Coding & Solution

Date	5 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation details of the HDI Prediction System. The application is developed using Python, Flask, and Machine Learning techniques to predict the Human Development Index based on socio-economic indicators. The solution follows a modular architecture, clean coding practices, and satisfies the functional requirements of the project.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Harishmagodey/HDI-Prediction-System-SmartBridge
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Data preprocessing, HDI prediction, Machine Learning model, Web interface, Result display
Pending / Incomplete Features	User authentication, Cloud deployment, Advanced analytics dashboard
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> and run <code>python app.py</code> .

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Prediction System is implemented with a clean and modular code structure. The application is easy to understand, maintain, and extend. GitHub is used for version control, and the project can be further enhanced by integrating additional datasets, improving model accuracy, and deploying the application on a cloud platform.